Problem Set 4

Important:

- Write your name as well as your NU ID on your assignment. Please number your problems.
- Submit both results and your code.
- Give complete answers. Do not just give the final answer; instead show steps you went through to get there and explain what you are doing. Do not leave out critical intermediate steps.
- This assignment must be submitted electronically through Gradescope by March 28th 2025 (Friday) by 11:59 PM.

1 Vector Norms

Consider the following vector:

$$\vec{x} = \begin{bmatrix}1.7 & 1.8 & 1.9 & 2.2 & 2.3 & 2.4 & 2.4 & 2.4 & 3.9\end{bmatrix}^T$$

i) (3 points) **What is $\|\vec{x}\|_1$? What is $\|\vec{x}\|_2$? What is $\|\vec{x}\|_\infty$?**

ii) (3 points) We now replace the outlier entry (last entry) with the average of the other entries, and get the following vector:

$$\vec{x} = \begin{bmatrix}1.7 & 1.8 & 1.9 & 2.2 & 2.3 & 2.4 & 2.4 & 2.4 & 2.1\end{bmatrix}^T.$$

Now, **what is $\|\vec{x}\|_1$? What is $\|\vec{x}\|_2$? What is $\|\vec{x}\|_\infty$?**
What was the relative change of **each** vector norm? That is, we need you to compute relative change as the difference of the value in part (i) and the value in part (ii) divided by the value in part(i), e.g.

$$\frac{|\|\vec{x}\|_{ii} - \|\vec{x}\|_i|}{\|\vec{x}\|_i}.$$

iii) (3 points) Which norm is least sensitive (most robust) to outliers? Is this the same as what you expected?

2 Vector and Matrix Calculus

You are given a vector $\vec{x} = \begin{pmatrix}x_1 \\ x_2\end{pmatrix}$, a vector $\vec{b} = \begin{pmatrix}b_1 \\ b_2\end{pmatrix}$ and a matrix $A = \begin{pmatrix}a_{11} & a_{12} \\ a_{21} & a_{22}\end{pmatrix}$. In this problem we will be deriving gradients (equivalent of a first derivative for vectors) and hessians (equivalent of a second derivative for vectors) of expressions that you will encounter often in Machine Learning models. These expressions are known as inner products and quadratic forms.

1. **Inner/dot products**: The inner/dot product of two vectors $\vec{x}$ and $\vec{b}$ can be denoted by $\vec{b}^T\vec{x}$.

   (a) Compute the inner product $\vec{b}^T\vec{x}$.

(b) Show that the derivatives of $\vec{b}^T\vec{x}$ with respect to x_1 and x_2, denoted by, respectively,

$$\frac{\partial \vec{b}^T\vec{x}}{\partial x_1} \text{ and } \frac{\partial \vec{b}^T\vec{x}}{\partial x_2},$$

are given by,

$$\frac{\partial \vec{b}^T\vec{x}}{\partial x_1} = b_1 \text{ and } \frac{\partial \vec{b}^T\vec{x}}{\partial x_2} = b_2. \tag{1}$$

These derivatives are known as partial derivatives given that $\vec{b}^T\vec{x}$ is a function of several variables.

(c) The gradient of $\vec{b}^T\vec{x}$, denoted by $\nabla_{\vec{x}}\ \vec{b}^T\vec{x}$ is the vector that is defined by,

$$\nabla_{\vec{x}}\ \vec{b}^T\vec{x} = \begin{pmatrix} \partial \vec{b}^T\vec{x}/\partial x_1 \\ \partial \vec{b}^T\vec{x}/\partial x_2 \end{pmatrix}. \tag{2}$$

What is $\nabla_{\vec{x}}\ \vec{b}^T\vec{x}$ equal to? Is this expected?

Hint: This is very similar to differentiating a linear function $f(x) = bx$ with respect to x.

2. **Quadratic forms**: A quadratic form of a vector $\vec{x}$ and matrix A is denoted by $\frac{1}{2}\vec{x}^T A\vec{x}$.

(a) Show that $\frac{1}{2}\vec{x}^T A\vec{x} = \frac{1}{2}\Big(a_{11}x_1^2 + (a_{12} + a_{21})x_1x_2 + a_{22}x_2^2\Big)$.

(b) Show that the derivatives of $\frac{1}{2}\vec{x}^T A\vec{x}$ with respect to x_1 and x_2 are given by,

$$\frac{\partial \frac{1}{2}\vec{x}^T A\vec{x}}{\partial x_1} = a_{11}x_1 + \frac{1}{2}(a_{12} + a_{21})x_2 \text{ and } \frac{\partial \frac{1}{2}\vec{x}^T A\vec{x}}{\partial x_2} = \frac{1}{2}(a_{12} + a_{21})x_1 + a_{22}x_2. \tag{3}$$

In the remaining parts, we assume that A is a symmetric matrix, that is $a_{12} = a_{21}$.

(c) The gradient of $\vec{x}^T A\vec{x}$ is defined by,

$$\nabla_{\vec{x}}\ \frac{1}{2}\vec{x}^T A\vec{x} = \begin{pmatrix} \partial \frac{1}{2}\vec{x}^T A\vec{x}/\partial x_1 \\ \partial \frac{1}{2}\vec{x}^T A\vec{x}/\partial x_2 \end{pmatrix}. \tag{4}$$

What is $\nabla_{\vec{x}}\ \frac{1}{2}\vec{x}^T A\vec{x}$ equal to? Is this expected?

(d) The hessian of $\frac{1}{2}\vec{x}^T A\vec{x}$ is defined by the matrix,

$$\nabla^2_{\vec{x}}\ \frac{1}{2}\vec{x}^T A\vec{x} = \begin{pmatrix} \partial^2 \frac{1}{2}\vec{x}^T A\vec{x}/\partial x_1^2 & \partial^2 \frac{1}{2}\vec{x}^T A\vec{x}/\partial x_1\partial x_2 \\ \partial^2 \frac{1}{2}\vec{x}^T A\vec{x}/\partial x_2\partial x_1 & \partial^2 \frac{1}{2}\vec{x}^T A\vec{x}/\partial x_2^2 \end{pmatrix}. \tag{5}$$

We now need to compute second derivatives of the quadratic form.

- Start by computing each component of the matrix $\nabla^2_{\vec{x}}\ \frac{1}{2}\vec{x}^T A\vec{x}$. Computing $\frac{\partial^2 \frac{1}{2}\vec{x}^T A\vec{x}}{\partial x_1^2}$ is equivalent to computing the derivative of $\frac{\partial \frac{1}{2}\vec{x}^T A\vec{x}}{\partial x_1}$ with respect to x_1. Similarly, to compute $\frac{\partial^2 \frac{1}{2}\vec{x}^T A\vec{x}}{\partial x_1 \partial x_2}$, you need to find the derivative of $\frac{\partial \frac{1}{2}\vec{x}^T A\vec{x}}{\partial x_1}$ with respect to x_2.
- Plug in all of these partial derivatives in the matrix. What do you notice $\frac{1}{2}\vec{x}^T A\vec{x}$ is equal to? Is that the same as in the scalar case?

3. **Weighted least squares**: Let $y_1, ..., y_n$ be real numbers representing positions on a number line. Let $w_1, ..., w_n$ be positive real numbers representing the importance of each of these positions. Consider the quadratic function, $f(\beta) = \frac{1}{2}\sum_{i=1}^n w_i(\beta - y_i)^2$. Note that β here is a scalar.

(a) Show that $f(\beta) = \frac{1}{2}\vec{x}^T W\vec{x}$ where $\vec{x} = \beta \cdot \vec{1} - \vec{y}$ and $W = \begin{pmatrix} w_1 & 0 & \dots & 0 \\ 0 & w_2 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & w_n \end{pmatrix}$. The $\vec{1}$ is the vector of all ones.

(b) Using part 2, find the value of β^* which minimizes $f(\beta)$ by solving $f'(\beta) = 0$.

(c) Show that the optimum is indeed a minimum by proving that $f''(\beta^*) > 0$.

3 Predicting Movie Ratings

Suppose that we are now interested in predicting a numeric rating for movie reviews. Given a movie review x, we define $\vec{\phi(x)}$ to be a vector of 0s and 1s which checks for the existence of specific words in the review x. We will use a non-linear predictor that takes a movie review x and returns $\sigma\big(\vec{w}^T\vec{\phi}(x)\big)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the logistic function that maps a real number to the range $(0, 1)$. For this problem, assume that the movie rating y is a real-valued variable in the range $[0, 1]$. Do not use math software such as Wolfram Alpha to solve this problem.

1. We use the squared loss function given by,

$$\mathcal{L}(x, y, \vec{w}) = \frac{1}{2}\Big(\sigma\big(\vec{w}^T\vec{\phi}(x)\big) - y\Big)^2, \tag{6}$$

for a single datapoint (x, y). Let $p = \sigma(\vec{w}^T\vec{\phi}(x))$ be the predicted value.

Show that the gradient of the loss $\mathcal{L}(x, y, \vec{w})$ with respect to $\vec{w}$, i.e. $\nabla_{\vec{w}}\mathcal{L}(x, y, \vec{w})$, is given by,

$$\nabla_{\vec{w}}\mathcal{L}(x, y, \vec{w}) = (p - y) \cdot p \cdot (1 - p) \cdot \vec{\phi}(x). \tag{7}$$

2. Suppose there is one datapoint (x, y) with some arbitrary $\phi(x)$ and $y = 1$. Specify conditions for $\vec{w}$ to make the magnitude of the gradient of the loss with respect to $\vec{w}$ arbitrarily small (i.e. minimize the magnitude of the gradient). Can the magnitude of the gradient with respect to $\vec{w}$ ever be exactly zero? You are allowed to make the magnitude of $\vec{w}$ arbitrarily large but not infinity.

4 Linear Regression Generalized

We have seen in class how to fit a linear function of the data for regression problems.

In this problem, we will explore how linear regression can also be used to fit non-linear functions of the data using feature maps. We will also discuss some the limitations of linear regression, for which future lectures will discuss solutions.

1. Polynomials of degree 3

Suppose we have a dataset $\mathcal{D} = \{(x^{(i)}, y^{(i)})\}_{i=1}^n$, where $x^{(i)}, y^{(i)} \in \mathbb{R}$. We want to fit a third degree polynomial $h_\theta(x) = \theta_3 x^3 + \theta_2 x^2 + \theta_1 x + \theta_0$, parametrized by $\vec{\theta} = \begin{pmatrix}\theta_0\\ \theta_1\\ \theta_2\\ \theta_3\end{pmatrix}$, to the dataset. You can see that the function $h_\theta(x)$ is still linear with respect to $\theta_0, \theta_1, \theta_2$ and θ_3.

We define $\phi(x)$ to be a function that transforms the original input x to a vector such that,

$$\phi(x) = \begin{pmatrix}1\\ x\\ x^2\\ x^3\end{pmatrix}. \tag{8}$$

For any datapoint of the form $(x^{(i)}, y^{(i)})$ with input $x^{(i)}$, we denote $\vec{\tilde{x}}^{(i)} = \phi(x^{(i)})$.

(a) Prove that $h_\theta(x) = \vec{\theta}^T\vec{\tilde{x}}$. This shows that the hypothesis function is a linear function with respect to $\vec{\tilde{x}}$ and linear regression can be applied on new dataset $\tilde{\mathcal{D}} = \{(\vec{\tilde{x}}^{(i)}, y^{(i)})\}_{i=1}^n$.

(b) Write the objective function $L(\vec{\theta})$ of the linear regression problem on the new dataset $\tilde{\mathcal{D}}$.

(c) Differentiate $L(\vec{\theta})$ with respect to $\vec{\theta}$. Show that, to find the optimal $\vec{\theta}$, you need to solve the following equations, known as the normal equations,

$$\tilde{X}^T\tilde{X}\vec{\theta} = \tilde{X}^T\vec{y}, \tag{9}$$

where $\tilde{X} = \begin{pmatrix} 1 & x^{(1)} & \left(x^{(1)}\right)^2 & \left(x^{(1)}\right)^3 \\ 1 & x^{(2)} & \left(x^{(2)}\right)^2 & \left(x^{(2)}\right)^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x^{(n)} & \left(x^{(n)}\right)^2 & \left(x^{(n)}\right)^3 \end{pmatrix} = \begin{pmatrix} \left(\vec{\tilde{x}}^{(1)}\right)^T \\ \left(\vec{\tilde{x}}^{(2)}\right)^T \\ \vdots \\ \left(\vec{\tilde{x}}^{(n)}\right)^T \end{pmatrix}$ and $\vec{y} = \begin{pmatrix} y^{(1)} \\ y^{(2)} \\ \vdots \\ y^{(n)} \end{pmatrix}$. The expressions you derived in Problem 2 will be very useful for this question.

(d) **Coding question:** We will now use the datasets provided on Canvas to build the linear regression model for polynomials of degree 3. The dataset we will use in this question is named,

`train.csv`

The data file contains two columns: x and y. As described in the introduction of this problem, x is the input and y is the output label.

Use the formulation of part (c) to find the optimal $\vec{\theta}$. You can use numpy. I am providing you with a starter code in which you have to fill some functions:

- Fill the function called *create_poly(self, k, X)*: The input given in this function is a $n \times 2$ matrix such that its first column is a column of 1s and the second column contains the values of $x^{(i)}$, for $i = 1, 2, ..., n$. Using the input X, form the matrix $\tilde{X}$ which will be the output of this function.
- Fill the function called *fit(self, X, y)*: Compute the matrix $\tilde{X}^T \tilde{X}$. The numpy function *np.dot()* will be helpful. Use *np.linalg.solve()* to solve the linear system given in equation 9 and update the value of *self.theta*. Do not compute the inverse of the matrix $\tilde{X}^T \tilde{X}$.
- Fill the function called *predict(self, X)*: In this function, you want to compute $h_\theta(x)$ for every example $x^{(i)}$.
- Fill in the function *run_exp()*: In this function, you will first need to create an instance of the class LinearModel(). Next, you will form the matrix $\tilde{X}$, update the value of $\vec{\theta}$ and compute the predicted $\vec{y}$, which you will name *plot_y*.
- In the main function, use the *run_exp()* to run the desired experiment. The provided code will, by itself, create a scatter plot of the training data, and plot the learnt hypothesis as a smooth curve over it.

2. Polynomials of degree k

We extend the idea above to polynomials of degree k by modifying the function ϕ such that,

$$\phi(x) = \begin{pmatrix} 1 \\ x \\ x^2 \\ \vdots \\ x^k \end{pmatrix}. \tag{10}$$

Follow the same procedure as in the previous sub-question, and implement the algorithm with $k = 3, 5, 10, 20$. You will obtain a similar plot as in the previous sub-question with hypothesis curves for each value of k. How is the fitted function performing for larger values of k?

3. Small datasets

For the rest of the problem, we will consider a small dataset with much fewer examples. The code is also provided on Canvas and named,

`small.csv`

We will be exploring what happens when the number of features start becoming bigger than the number of examples in the training set. Run your algorithm on this small dataset using the function $\phi(x)$ in 10 with $k = 1, 2, 5, 10, 20$. Create a plot of the various hypothesis curves, just like previous sub-questions. What do you notice about the behavior of the fitting functions as k increases? Does increasing k always yield better results?